CP–6232

Sub. Code
11

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

TECHNICAL ENGLISH

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Add prefix to the following words :

   (a) Normal

   (b) Final

   (c) Legal.

2. Write the Synonyms of the following words :

   (a) Certain

   (b) Success

   (c) Plenty.

3. Rewrite as directed :

   (a) We – (go) to meet the CM yesterdays

   (supply the correct tense of the word)

   (b) The movement of the customers is watched by the security.

4. What do you mean by "transfer of information"?

5. What is the purpose of message in an organisation?

6. Describe the functions of an Ipad.

7. Write down the comparative adjective of 'bad' and 'good'.

8. What is a 'bio-data'?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write an essay on the different kinds of Reading skills.

Or

(b) Write short notes on any FOUR of the following :

(i) comparative adjective

(ii) non-verbal communication

(iii) sequencing of sentences

(iv) classification of letters

(v) planning the visuals in presentation.

10. (a) What are the kinds of prepositions? Explain with illustrations.

Or

(b) Discuss "Interpersonal relationship".

11. (a) Write a paragraph on your pet animal.

Or

(b) Send an e-mail to a leading travel agency asking information on various package tours offered by them.

12. (a) Write a letter of complaint to a newspaper exposing the misappropriation of funds in an orphanage of your locality.

Or

(b) Apply for the post of supervisor in Public works Department with your bio-data.

Wk 9

13. (a) What are the six types of Presentations? Explain with examples.

Or

(b) Write down a Mock–interview between the manager of a company and a candidate appearing for a clerical post?

CP–6233

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12

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

MATHEMATICS — I

Time : 3 Hours Maximum : 75 Marks

Part A $(5 \times 3 = 15)$

Answer any **five** questions.

1. Find the rank of the matrix $A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & -4 & -2 \\ -1 & 2 & 1 \end{bmatrix}$.

2. Calculate A^3 when $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ using characteristic equation.

3. Prove that the points $(3,-1,1), (5,-4,2)$ and $(11, -13, 5)$ are collinear by direction cosines.

4. Find the point where the line joining the points $(2, -3, 1)$ and $(3,-4, 5)$ cuts the plane $2x + y + z = 7$.

5. Find the radius of curvature to the curve $y^2 = 12x$ at $(3, 6)$.

6. Find the centre of curvature at $(3, 27)$ on the curve $y = x^3$.

7. If $u = xy + yz + zx$ whose $x = \frac{1}{t}$, $y = e^t$, $z = e^{-t}$ find $\frac{du}{dt}$.

8. Solve $\left(D^2 + 4D + 4\right)y = e^{2x}$.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 1 \\ -7 & 2 & -3 \end{bmatrix}$.

Or

(b) Examine by using the theory of matrices whether the following equations are consistent. Solve when consistent, $x + y + z = 6$, $2x + y + 3z = 13$, $5x + 2y + z = 12$.

10. (a) Show that the lines $\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}$ and $\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$ are coplanar and find their common point.

Or

(b) Prove that the two sphere $x^2 + y^2 + z^2 - 2x + 4y - 4z = 0$ and $x^2 + y^2 + z^2 + 10x + 2z + 10 = 0$ touch each other and find the co-ordinates of the point of contact.

11. (a) Show that ρ at any point $\left(a\cos^3\theta, a\sin^3\theta\right)$ on the curve $x^{2/3}+y^{2/3}=a^{2/3}$ is $3a\sin\theta.\cos\theta$.

Or

(b) Show that the centre of curvature at any point (x, y) on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\left(\frac{a^2-b^2}{a^4}x^2, \frac{a^2-b^2}{b^4}y^2\right)$.

12. (a) If $z=f(x,y)$ where $x=e^u\cos\theta$, $y=e^u\sin\theta$, show that $y\frac{\partial z}{\partial u}+x\frac{\partial z}{\partial \theta}=e^{2u}\frac{\partial z}{\partial y}$.

Or

(b) Investigate the maxima of the function $f(x, y)=x^3y^2(1-x-y)$.

13. (a) Solve $x^2\frac{d^2y}{dx^2}+4x\frac{dy}{dx}+2y=\sin(\log x)$.

Or

(b) Solve $\frac{d^2y}{dx^2}+4y=4\tan 2x$ using method of variation of parameters.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

PRODUCTION TECHNOLOGY

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What are the various operations performed by a Lathe?

2. Define : Single point cutting tool.

3. Mention any two differences between Drilling and Reaming.

4. Discuss the three types of flames used in gas welding.

5. List the sheet metal operations and explain any two operations.

6. Mention the various types of gears and its applications.

7. What is the purpose of flux used in the welding process?

8. Define : Limits, Fits and Tolerances.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe different types of Lathes and explain the various parts of a Lathe with diagram.

Or

(b) Explain the feed and thread cutting mechanism of a Lathe with a neat diagram.

10. (a) What are the safety precautions to be taken for various processes. Explain.

Or

(b) Explain in detail :

(i) Welding

(ii) Soldering

(iii) Brazing and

(iv) Bronze welding.

11. (a) What are the different types of tools used in sheet metal work? Explain with suitable sketches.

Or

(b) Explain the principle of sheet metal work and calculation of rivetted joints.

12. (a) Explain in detail the working of any one type of milling machine with neat sketch and its uses.

Or

(b) Explain the different types of bearings and their functions in detail.

13. (a) Explain briefly the various precision measuring instruments used in aeronautical industry with simple diagrams.

Or

(b) Explain the tumbler gear mechanism in detail with neat sketch.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

ENGINEERING MECHANICS AND STRENGTH OF MATERIALS

Time : 3 Hours Maximum : 75 Marks

Part A $(5 \times 3 = 15)$

Answer any **five** questions.

1. Define the following terms :

   (a) Coplanar forces

   (b) Concurrent forces.

2. State the necessary and sufficient conditions for static equilibrium of a particle in two dimensions.

3. When will the product of inertia of a lamina become zero?

4. A body moves along a straight line so that its displacement from a fixed point on the line is given by $s = 3t^2 + 2t$. Find the velocity and acceleration at the end of 3 sec.

5. State the laws of dry friction.

6. What is general plane motion?

7. Define free and forced vibrations.

8. Define parallelogram law of forces.

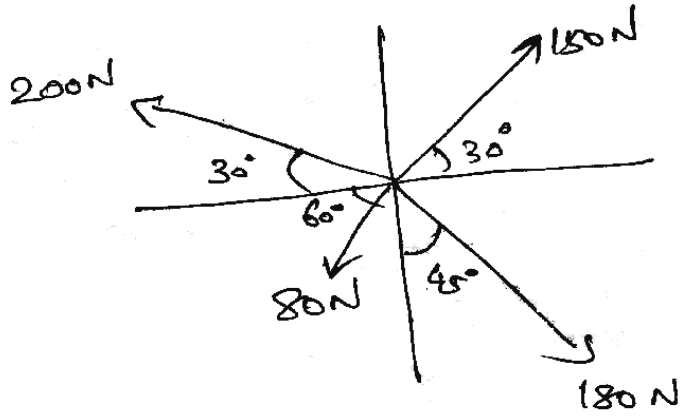
Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the fundamental units and derived units.

Or

(b) Determine the resultant of the concurrent force system.



10. (a) Explain the resolution of force of two components on a system.

Or

(b) Explain the procedure of finding the resultant force of more than two concurrent forces.

11. (a) Explain briefly about angle of friction with neat sketch.

Or

(b) Explain the followings :

(i) Screws

(ii) Wedges

(iii) Thrust bearing.

12. (a) Explain briefly about deficient and redundant frames.

Or

(b) What are the various types of Trusses? Explain any one briefly.

13. (a) What are the various types of loads? Explain any two with bending moment diagram.

Or

(b) Explain the Eulers theory.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

MATHEMATICS – II

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Evaluate $\int_2^a \int_2^b \frac{dx\,dy}{xy}$.

2. If S is any closed surface enclosing a volume V and $\vec{F} = ax\vec{i} + by\vec{j} + cz\vec{k}$ prove that $\iint_S \vec{F}.\hat{n}\,dS = (a+b+c)V$.

3. If $u + iv$ is analytic, show that $v - iu$ is also analytic.

4. For what values of a, b and c the function $f(z) = x - 2ay + i(bs - cy)$ is analytic.

5. Evaluate $\int_c \frac{dz}{z+4}$ where c is the circle $|z| = 2$.

6. Evaluate $\int_c \frac{e^z}{z+1} dz$, where c is the circle $\left|z + \frac{1}{2}\right| = 1$.

7. Find the Laplace transform of $e^{-2t}\cos^2 t$.

8. Find the inverse Laplace transform of $\frac{1}{(s+a)^n}$, where n is a positive integer.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Change the order of integration in $\int_0^a \int_{x^2/a}^{2a-x} xy\,dy\,dx$ and hence evaluate it.

Or

(b) Find the volume of that portion of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ which lies in the first octant.

10. (a) Using Green's theorem in the plane, evaluate $\int_c (3x^2-8y^2)\,dx+(4y-6xy)\,dy$, where c is the boundary of the region enclosed by $y=\sqrt{x}$ and $y=x^2$.

Or

(b) Verify Stoke's theorem for $\vec{F}=(2x-y)\vec{i}-yz^2\vec{j}-y^2z\vec{k}$, where S is the upper half of the sphere $x^2+y^2+z^2=1$, and c is its boundary.

11. (a) (i) Prove that the functions $v=e^{-x}(x\cos y+y\sin y)$ is harmonic and determine the corresponding analytic function $f(z)$.

(ii) Determine the conformal representation of $W=z^2$.

Or

(b) (i) If $f(z)$ is an analytic function of z prove that $\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f(z)|^2$.

(ii) Under the transformation $W = \frac{1}{z}$ find the image of $|z-2i| = 2$.

12. (a) Using method of contour integration, evaluate $\int_0^\infty \frac{x^2}{(x^2+1)(x^2+4)}dx$.

Or

(b) Evaluate $\int_0^{2\pi} \frac{\sin^2\theta}{5-4\cos\theta}d\theta$ using contour integration.

13. (a) Evaluate $L^{-1}\left[\frac{3s^2+16s+26}{s(s^2+4s+13)}\right]$.

Or

(b) Solve $y''+y = \sin t$, given $y(0)=1, y'(0)=\frac{1}{2}$ using Laplace transform.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

FLUID MECHANICS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define Surface tension with a sketch.
2. Define mechanical gauge and mention its types.
3. Define the term 'Centre of Buoyancy'.
4. Mention the types of fluid flow.
5. Write the Bernoulli's equation in real fluid.
6. Distinguish between orifice meter and pitot tube in fluid flow.
7. Differentiate laminar flow and turbine flow.
8. Define Magnus effect.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain any five properties of fluids.

Or

(b) Explain the working principle of
   (i) Piezometer
   (ii) Hydraulic accumulator.

10. (a) Explain the conditions for equillibrium stability of Suberged body.

Or

(b) The right limb of a simple U-tube manometer containing (Hg) is open to the atmosphere while the left limb is connected to a pile in which a fluid of sp-gravity 0.9 is flowing. The centre of the pipe is 12 cm below the level of mercury in the right limb. Find the pressure.

11. (a) Obtain an expression for the

(i) velocity potential function

(ii) discharge and continuity equation.

Or

(b) Discuss :

(i) source flow

(ii) sink flow.

12. (a) Derive the Euler's equation of motion.

Or

(b) Derive the equation for discharge through venturimeter.

13. (a) Explain the types of boundary layer (any four only).

Or

(b) Discuss the expression for

(i) drag and lift

(ii) bluff body.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

THERMODYNAMICS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What do you mean by State?
2. Explain the term Entropy.
3. State and explain Joule's law
4. Why C_p is greater than C_v ?
5. Distinguish between ideal gas and real gas.
6. What is Dalton's law of partial pressure?
7. Briefly explain efficiency of a compressor.
8. Write a short note on turbine.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the following :
   (i) Thermodynamic equilibrium
   (ii) System and surroundings.

   Or

   (b) Discuss about steady flow and derive an expression for efficiency of it.

10. (a) State Boyle's law and Charles law and derive an equation for the state of a perfect gas.

Or

(b) Describe a Otto cycle with necessary diagram and obtain efficiency equation.

11. (a) Explain Enthalpy. Discuss the account of specific heat of gas mixtures.

Or

(b) (i) Explain theoretical air and excess air.

(ii) Explain the determination of measuring air required for combustion.

12. (a) Describe in detail the working of reciprocating compressor.

Or

(b) Explain the working of axial flow compressor with necessary diagram.

13. (a) Describe the working of constant pressure closed cycle gas turbine with its merits and demerits.

Or

(b) Explain the following briefly :

(i) Thrust power

(ii) Propulsion efficiency

(iii) Rocket propulsion

(iv) Thermal efficiency.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

BASIC ELECTRICITY AND ELECTRONICS

Time : 3 Hours Maximum : 75 Marks

Section A (5 × 3 = 15)

Answer any **five** questions.

1. State Thevenin's theorem.
2. What is a dc generator?
3. Explain electromagnetic induction.
4. Define capacity of a capacitor. Give its units.
5. Define rectification efficiency.
6. Give the difference between current and voltage transformer.
7. Mention some applications of function generators.
8. List the characteristics of a Bridge rectifier.

Section B (5 × 12 = 60)

Answer **all** questions.

9. (a) (i) State and explain Maxwell's superposition theorem.

(ii) State and explain Ohm's law.

Or

(b) Obtain expressions for the value of equivalent resistances when three resistors R_1, R_2 and R_3 are connected in

(i) Series

(ii) Parallel.

10. (a) Describe the construction and operation of a transformer. Obtain the emf equation.

Or

(b) Explain the construction and working of a DC generator. Derive the emf equation for it.

11. (a) With a neat diagram, describe the construction and operation of a induction motor. Obtain the torque equation.

Or

(b) What is a single phase induction motor? Explain its construction and principle of operation, with a neat sketch.

12. (a) What is a UJT? Explain the construction, opration and characteristics of UJT.

Or

(b) With necessary circuit diagram, explain the construction and working of a multimeter.

13. (a) Describe the construction, working of a full wave rectifier with a neat diagram. Also obtain the value of rectification efficiency.

Or

(b) Give an account of

(i) Voltage regulator

(ii) SMPS.

CP–6240

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

MECHANICS OF FLIGHT

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write the significance of Troposphere?

2. What is the importance of relative density in atmosphere?

3. Write the basic concept of boundary layer.

4. Define aspect ratio and its importance.

5. Write role of spoiler in aircraft.

6. Write the thrust required Vs thrust available curve and its significance.

7. Define the take-off distance.

8. What is weathercock stability?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe the various layers of atmosphere and its individual characters.

Or

(b) Write notes on :

(i) Newton's law of motion.

(ii) Venturi Effect.

10. (a) Write notes on aerofoil terminology.

Or

(b) Explain about Reynolds number Vs velocity, profile drag Vs velocity, total drag Vs velocity.

11. (a) Explain the airfoil characteristics and its symbols.

Or

(b) Describe the plan form geometry with sketch and explain the associated terms.

12. (a) Explain about the three axes of an aircraft with neat sketch.

Or

(b) Describe the moveable aircraft controls and their functions with neat sketch.

13. (a) List out and explain the functions of take-off performance.

Or

(b) Explain the directional stability and their control.

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CP–6241

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32

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT MATERIALS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write the importance of aluminium code.

2. What are the disadvantages of stainless steel?

3. What is the significance of Brinell hardness test?

4. What is the importance of hardening of metal?

5. What is silicon resin?

6. List out the defects in wood.

7. Define the term heat treatment.

8. What is known as reinforcing material?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write brief notes on magnesium alloy.

Or

(b) List out the properties of titanium.

10. (a) Explain the Rockwell Hardness test.

Or

(b) Discuss about the eddy current inspection.

11. (a) What is hardening process and explain the case hardening process?

Or

(b) Briefly explain the electroplating process.

12. (a) Briefly discuss about the solid wood.

Or

(b) Explain the nut designation codes.

13. (a) List out the advantages and disadvantages of composites.

Or

(b) Write notes on thermosetting resins.

CP–6242

Sub. Code
33

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT STRUCTURES

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write the aircraft zoning numbers.

2. What are the advantages of semi-monocoque fuselage construction?

3. Write short notes on power flying control unit.

4. Write the basic principles of fly-by-wire system and its components.

5. What is torque link, side strut and its purposes?

6. What is overall weight and its importance?

7. What is levelling and its procedure?

8. Define dihedral angle and its significance.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain about the Sandwich construction.

Or

(b) Write short notes on :

(i) Fail Safe Design

(ii) Mid wing aircraft.

10. (a) Explain the constructional features of flaps, leading edge slots and its types.

Or

(b) Explain the description of power assisted controls of flight controls.

11. (a) Explain about the Power Boost brake system of transport aircraft.

Or

(b) Briefly explain about the nose wheel steering system.

12. (a) Explain the aircraft weighing procedure of small executive aircraft.

Or

(b) Write notes on weight and balance data.

13. (a) Describe the procedure of Levelling of any transport aircraft.

Or

(b) Explain the Rigging of flight controls.

CP–6243

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT SYSTEMS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is the purpose of seals and name their classifications?

2. Define Pascal's law.

3. Explain the functioning of pneumatic check value.

4. What is the purpose of a cold air unit?

5. Name the five basic requirements for cabin pressurization and air conditioning system.

6. List down the methods used for control of ice formation in aircraft.

7. What is volatility of fuel?

8. Why fuel Jettisoning is alone?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) What are the purpose of hydraulic accumulator and describe its type with figure?

Or

(b) Enumerate the types of fluid contaminations and explain the various contamination checks.

10. (a) List down the advantages and disadvantages of pneumatic system over hydraulic system.

Or

(b) Write short notes on :

(i) Pressure regulator

(ii) Pneumatic control value.

11. (a) Explain the Aircycle cooling system of Aircraft air conditioning system.

Or

(b) Why it is necessary to pressurize the aircraft cabin?

12. (a) Explain the operation of pneumatic deicing system in detail.

Or

(b) Explain wind shield rain repellent system.

13. (a) Describe with diagram, the light aircraft gravity fuel feed system.

Or

(b) Describe single point fueling.

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CP–6244

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41

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

PISTON ENGINE AND PROPELLER

Time : 3 Hours Maximum : 75 Marks

Part A ($5 \times 3 = 15$)

Answer any **five** questions.

1. Define Boyles law and Charles law.

2. Define compression ratio.

3. What is the purpose of Crankshaft? Name the different parts of Crankshaft.

4. What are the differences between internal and external superchargers?

5. Define :

   (a) Cloud point

   (b) Pourpoint of lubricating oil.

6. Write the advantages of pressure injection carburetor over float carburetor.

7. Define :

   (a) Geometric pitch of propeller

   (b) Effective pitch of propeller.

8. What do you understand by impulse coupling?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the operation of Otto cycle with P.V diagram.

Or

(b) Explain about the various factors which affects engine performance.

10. (a) Explain the construction detail of different connecting rod assemblies with neat sketch.

Or

(b) Describe the simple induction system with diagram.

11. (a) Name the various components of float carburetor and explain about main metering system and Idling system in detail.

Or

(b) Explain the operation of dry sump lubrication system with diagram.

12. (a) Briefly explain about the operation of constant speed propeller.

Or

(b) Describe about various forces acting on propeller with diagram.

13. (a) What are the primary circuit components of magneto ignition system and explain about their operations?

Or

(b) Describe the construction detail of spark plug.

CP–6245

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

GAS TURBINE ENGINE

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write short notes on super sonic inlet duets of gas turbine engine.

2. List out the factors that affect the amount of thrust produced by a propeller blade.

3. Explain how anti-icing is performed on inlet guide vanes.

4. What is rich blow out and lean die out?

5. Define the term ESHP. Give its significance.

6. Differentiate 'Gross thrust' and 'Net thrust'.

7. Explain the blade element theory of propeller.

8. What is called power turbine?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain in detail the principles of jet propulsion.

Or

(b) Derive the thrust equations for a gas turbine engine.

10. (a) Describe the working of axial flow compressor in detail.

Or

(b) Describe 'thrust augmentation'. Explain about various methods used for thrust augmentation in gas turbine engine.

11. (a) Explain the forces acting on a propeller.

Or

(b) Describe the construction features of a turbo-prop engine.

12. (a) Explain Supervisory Electronic Engine Control (EEC).

Or

(b) Describe the types of fuel used in gas turbine engine and list the qualities of fuel for gas turbine engine.

13. (a) Describe the function of air turbine starter.

Or

(b) Explain high energy ignition system of a gas turbine engine.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT INSTRUMENTS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer **all** questions.

1. Write the types of instrument panel layout.

2. Write the effects of insufficient oxygen in the aircraft.

3. Write the basic uses of altimeter.

4. Write the principle of operation of directional gyro.

5. What is thermocouple and its application?

6. Write the types of air temperature gauges.

7. What is the significance of magnetic compensation?

8. Define magnetism.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write the notes on "description of operational range markings of airspeed indicator?

Or

(b) List out the installation procedure of oxygen system.

10. (a) Explain the Pitot-Static system used in Boeing-747 aircraft.

Or

(b) Describe the Q code setting procedure of altimeter.

11. (a) Write short notes on Artificial Horizon.

Or

(b) Explain the procedure for error compensation in gyroscopic instruments.

12. (a) Describe the principle and operation of Exhaust Gas Temperature indicator.

Or

(b) Write short notes on:

(i) Manifold pressure gauge

(ii) Oil pressure gauge.

13. (a) Explain the principle of Remote Reading Compass system.

Or

(b) List out and describe the occasions for compass swinging.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT ELECTRICAL SYSTEM

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define Ohm's Law and write the various types of Electrical Loads?

2. Name the different types of Circuit Protection devices?

3. What do you mean by thermal runway and how to eliminate it?

4. Write few points about the Care and Maintenance of aircraft batteries?

5. Write the components of DC generator?

6. What do you mean by Armature Reaction and its methods of elimination?

7. Write the special features to be adopted in selection of aircraft cables?

8. Write the principle of Voltage Regulator and name the various types?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the construction features of atom and its structure with a neat diagram?

Or

(b) Explain the various methods of Producing Voltage?

10. (a) Briefly explain the construction feature of Lead Acid battery and its chemical action?

Or

(b) Write the precautions to be observed while handling the aircraft batteries and explain the charging procedure of it?

11. (a) Explain Electrical conduit and write the advantages of Crimping over Soldering?

Or

(b) Explain the operation of Carbon Pile Voltage regulator with a neat diagram?

12. (a) Write the procedure of carrying out Continuity test and Insulation test?

Or

(b) Explain briefly about Parallel and Split Bus bar System?

13. (a) Explain the Turbine Engine Starter Generator system and explain its operation?

Or

(b) Explain briefly about the External and Internal lighting of an aircraft?

CP–6248

Sub. Code
51

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT RULES AND AIRWORTHINESS REGULATIONS –I

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Discuss in brief about "Concessions" as applicable to aircraft maintenance.

2. What do you know about "Change of ownership" with reference to aircraft registration?

3. Write in brief about registration of aircraft in "Category A".

4. Discuss in brief about "Release Notes" and "Rejection Note".

5. What do you mean by scope of approval? Discuss in brief.

6. Write short notes on Quality Assurance.

7. Who issues certificate of maintenance? At what occasions this is issued.

8. Discuss in brief about aircraft rules and regulations, 1937.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write in detail about classification of Aircraft.

Or

(b) List out the documents that are submitted to DGCA for registration of Aircraft.

10. (a) Discuss in detail about Journey Log Book.

Or

(b) Write in detail about the requirements that are to be fulfilled for the issue of certificate of airworthiness.

11. (a) What do you mean by certifying staff ? Discuss in detail.

Or

(b) Discuss in detail about fuelling procedure of an aircraft.

12. (a) Explain in detail about type certification of an aircraft.

Or

(b) List out the duties and responsibilities of Accountable Manager.

13. (a) Discuss in detail about fixation of nationality and registration marks.

Or

(b) Explain in detail about mandatory modifications.

CP–6249

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52

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT MAINTENANCE

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write the significance of progressive inspection?

2. What is the significance of operational inspections of engine?

3. What is called Exfoliation?

4. Name the various fixed landing gear?

5. How will you prevent heat generated by braking action?

6. List out the effect of burst tyre incident?

7. What is template?

8. What is scriber and its importance?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Briefly explain the frequency and details of the Progressive Inspection?

Or

(b) Write notes on maintenance records?

10. (a) List out the various inspections of the aircraft is subjected to heavy/overweight landings?

Or

(b) List out the some of the principals involved for protection against corrosion.

11. (a) Explain the typical wheel removal procedure?

Or

(b) Write the importance of retraction test and its procedure?

12. (a) Write notes on rivet spacing and rivet size?

Or

(b) List out the criteria for selecting the sheet metal for any work?

13. (a) Briefly plan the template of Pre-flight inspection procedure?

Or

(b) Describe the procedures for cementing plastics?

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CP–6250

Sub. Code
53

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AERO ENGINE MAINTENANCE

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define overhaul and its types?

2. What is poppet valve and its construction details?

3. Write the importance of propeller balancing?

4. List out some precautions before the ground run of piston engine?

5. Define compressor wash and its importance?

6. Write the advantages of annular combustion chamber?

7. Define EGT and how to measure?

8. Write the necessasity of initial warm up of turbine engine?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe the compression testing of aircraft engine cylinders?

Or

(b) Briefly explain about the dimensional inspection of engine?

10. (a) Describe about the dynamic balancing of propeller?

Or

(b) List out the general inspection and repair of metal propellers?

11. (a) Briefly list out the details of inspection of ground run of reciprocating engine?

Or

(b) Explain the inspections of magnetos of piston engine ignition system?

12. (a) List out the non-routine inspections of gas turbine engine?

Or

(b) Describe the procedure of maintenance of fan blades and compressor blades?

13. (a) Describe the operational checks of the gas turbine engine?

Or

(b) How will you assess the engine performance in terms of engine pressure ratio parameter?

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CP–6251

Sub. Code
54

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT COMMUNICATION AND NAVIGATION SYSTEM

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Explain wavex length.
2. What is the difference between Radiofrequency and Audio frequency?
3. Describe the operation of hight – pass filter.
4. Describe Radio Magnetic Indication(RMI).
5. What is the purpose of Radio Tuning System.
6. What frequencies are used by ELT?
7. Describe the procedure for testing communication Radio.
8. What is Radome?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain superheterodyne receiver with the help of a block diagram.

Or

(b) Explain the transmitter of VHF communication system with the help of a block diagram.

10. (a) Explain the principle of operation of ADF.

Or

(b) Explain the operation of VOR receiver.

11. (a) Describe the operation of MLS.

Or

(b) Write notes on :

(i) Selcal decoder

(ii) HF communication system.

12. (a) Explain Global positioning system.

Or

(b) Explain Doppler navigation system.

13. (a) Describe the operation of analog Radar.

Or

(b) Write notes on :

(i) Radar safety

(ii) Wave guide.

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CP–6252

Sub. Code
61

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRCRAFT RULES AND AIRWORTHINESS REGULATIONS – II

Time : 3 Hours Maximum : 75 Marks

Part A ($5 \times 3 = 15$)

Answer any **five** questions.

1. Briefly discuss about the general requirements of a physician's kit.

2. What do you know about special flight permit. Briefly explain.

3. List out the documents that are related to aircraft maintenance.

4. Discuss briefly about the qualifications required by a pilot who carry out test flight of gliders.

5. Discuss in brief about GPWS (Ground Proximity Warning System).

6. State the functional requirment of Acas I and Acas II.

7. List out the preservation periods of different types of log book.

8. Differentiate between aircraft accident and aircraft indicent.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) State the procedures followed for reporting and investigation of accidents.

Or

(b) Explain in detail about "Minimum equipment list".

10. (a) Discuss in detail about aircraft flight manual.

Or

(b) Explain the procedure of obtaining authorisation and approval.

11. (a) Explain in detail the requirements of Emergency Locator Transmitter (ELT).

Or

(b) Discuss in detail about the aircraft operating in IFR and VFR condition.

12. (a) Discuss in detail about the procedure followed in the issuance of car.

Or

(b) Explain about various types of FDR and Tabulate the recording time against each type.

13. (a) Explain the procedure for subsequent revalidation of certificate of airworthiness.

Or

(b) Discuss in detail about aeromobile licence and RTR licence issued to maintenance/operations personnel.

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Sub. Code
62

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

INDUSTRIAL MANAGEMENT

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define management by objective.

2. What do you mean by authority?

3. What is manpower?

4. Define Recruitment.

5. Write any three duties of a supervisor.

6. What do you mean by motivation?

7. What is meant by office management?

8. Define inventory management.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) What is management? Explain the function of management.

Or

(b) Discuss in detail about MBO.

10. (a) What is an organization? Explain the various types of organizational structure.

Or

(b) Explain the steps in manpower planning.

11. (a) Discuss the importance and methods of motivation.

Or

(b) Explain the role and duties of a supervisor.

12. (a) Explain the techniques and ways of time management.

Or

(b) Discuss in detail about production management and its importance in an organisation.

13. (a) How will you manage and control an inventory?

Or

(b) Explain the purpose and procedure of work study.

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CP–6254

Sub. Code
63

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

AIRPORT AND AIRTRAFFIC SERVICES

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Discuss in brief about taxiway markings.

2. State the purpose of airtraffic control.

3. List out the factors that influence the required runway length.

4. What do you mean by runway incursions?

5. List out the goods that are prohibited from carrying in air.

6. Discuss in brief about the different aircraft parking configuration with neat sketch.

7. Discuss in brief about the significance of visibility.

8. Briefly discuss regarding the use of :

   (a) Marker beacon

   (b) DME.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain in detail about the runway configurations.

Or

(b) Elaborate the types of atc facilities provided at the airport.

10. (a) Explain the primary considerations for the layout of an airport.

Or

(b) Describe in detail about the various systems employed at airport security.

11. (a) Explain in detail about the operation of Instrument Landing System (ILS).

Or

(b) Explain in detail about the use of VOR, NDB and X-ray unit.

12. (a) Explain in detail about Apron, Holding Apron and terminal Apron.

Or

(b) Explain in detail about the effects of noise in airport.

13. (a) Explain in detail about the precession approach radar.

Or

(b) Discuss in detail about briefing centre and its functions.

CP–6255

Sub. Code
64

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aeronautical Science

GROUND HANDLING, SAFETY AND SUPPORT EQUIPMENT

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Explain various gas type fire extinguishing agents.

2. Explain NFPA standard #10 regarding marking of fire extinguishers.

3. Describe maintenance safety on aircraft tire mounting.

4. Describe signal lights shown from control tower to taxing aircraft.

5. Describe water contamination of fuel.

6. Explain the purpose and precautions of a single base jack.

7. What is the purpose of a electrical power unit?

8. Why a signal man is required when aircraft is taxying?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe the aircraft jacking procedure.

Or

(b) Briefly explain flight line safety.

10. (a) Describe the precautions against wind storm damage.

Or

(b) Briefly explain the cold weather protection of aircraft.

11. (a) Describe the maintenance safety of welding.

Or

(b) Describe the maintenance safety on compressed air system.

12. (a) Briefly explain the fueling operations.

Or

(b) Describe the duties of refueling crew.

13. (a) Explain the operating procedure and maintenance of ground air strat unit.

Or

(b) (i) Describe air conditioning and heating unit. (8)

(ii) Describe pre oiling equipment. (4)